

IT 4560 · Week 03 Lab Submission

Work Smarter: Systems Manager and S3 Static Website

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SECTION 2: REFLECTION AND ANALYSIS

RESPONSE 1 -- Systems Manager Feature for Automating Weekly Maintenance

The AWS Systems Manager Run Command capability is the best fit for Dana's task because it allows HarborTech to run the same maintenance on multiple EC2 instances without having to connect to each instance individually. Before the automation can work, the five EC2 instances must have these prerequisites: SSM Agent must be installed and running on each instance so Systems Manager can communicate with them, The instances must have the proper IAM permissions, usually through an IAM instance role, allowing Systems Manager to manage the instances, The instances must have connectivity to Systems Manager endpoints so they can communicate with the AWS service. HarborTech should verify that all five instances are managed nodes in Systems Manager before running the maintenance command.

RESPONSE 2 -- Parameter Store and Centralizing Application Configuration

Bright Path could use AWS Systems Manager Parameter Store to store the environment-specific value in one central location, such as /BrightPath/App/DatabaseEndpoint. Instead of having the database endpoint hardcoded in several files, the application or automation would retrieve the parameter by its name when it needs the value. The application or automation would need to be changed so it calls Parameter Store, such as using the AWS SDK or AWS CLI, and requests the parameter by its name. It would then use the returned value instead of relying on the hardcoded value. Parameter Store does not automatically rewrite existing files. It only stores and provides the centralized value. The existing configuration files would still contain the original hardcoded values unless HarborTech specifically changes the application, automation, or configuration process to retrieve the parameter and update or replace those values.

RESPONSE 3 -- S3 Static Website Hosting and Workload Fit

S3 static website hosting is an appropriate design for Bright Path because the page only needs to provide HTML, program hours, contact information, images, and downloadable public forms. These are static files that S3 can host without needing a web server or database. One future requirement that would make S3 static hosting insufficient is a donor portal that requires users to log in and access protected information. That would require server-side processing, authentication, and possibly a database, so Bright Path would need a different or expanded architecture.

SECTION 3: LEARNER LAB INVESTIGATION DOCUMENTATION

S3 WEBSITE CONFIGURATION, CLI EVIDENCE, SANDBOX RESULTS, AND SYSTEMS MANAGER RECOMMENDATION

I created a static webpage for Bright Path Community Services using an index.html file. The page includes the program hours, contact information, description of a Bright Path service, and information about public forms. The page does not contain real personal or confidential information.

S3 Bucket Information

Bucket Name: brightpath-as-39483

Region: us-west-2

Object Key: index.html

I uploaded the index.html file to the S3 bucket and enabled Static website hosting with index.html set as the index document.

Website Endpoint: <http://brightpath-as-39483.s3-website-us-east-1.amazonaws.com/>

Website Access Test

When I tested the S3 website endpoint

Access was denied, the Learner Lab restricted the public-access configuration needed for the website to be publicly readable. I documented this instead of attempting to bypass the sandbox controls. In a normal AWS account, the appropriate public-access configuration and permissions would be allowed for the website to be publicly accessible.

CLI Verification and Update

I first verified my AWS identity in CloudShell using:

```
aws sts get-caller-identity
```

The command returned my current AWS identity and confirmed that I was operating in the Learner Lab environment.

I then used the following command to upload or update the website files:

```
aws s3 sync ./brightpath-site s3://[YOUR-BUCKET-NAME]/
```

HarborTech Recommendation for Dana

Dana's repeated maintenance task across five EC2 instances is best handled with AWS Systems Manager Run Command. Run Command allows HarborTech to execute the same maintenance command consistently across multiple managed EC2 instances without connecting to each instance individually.

Before using Run Command, HarborTech should verify that the five EC2 instances are properly managed by Systems Manager. The SSM Agent must be installed and running, and the instances must have the appropriate IAM permissions through an instance role so Systems Manager can manage them. Instances also need connectivity to the required Systems Manager endpoints.

HarborTech should verify that all five instances appear as managed nodes before running the maintenance command. If an instance is not ready, it should be documented and corrected before relying on the automation.

SECTION 4: HARBORTECH OPERATIONS PLAYBOOK ENTRY

GITHUB PLAYBOOK ENTRY URL

<https://github.com/Kiyahbee205/week3/blob/main/README.md>

PLAYBOOK CONFIRMATION

Aakiyah Smith confirms that week03-systems-manager-s3.md has been published to a public GitHub repository and is accessible at the URL above.